

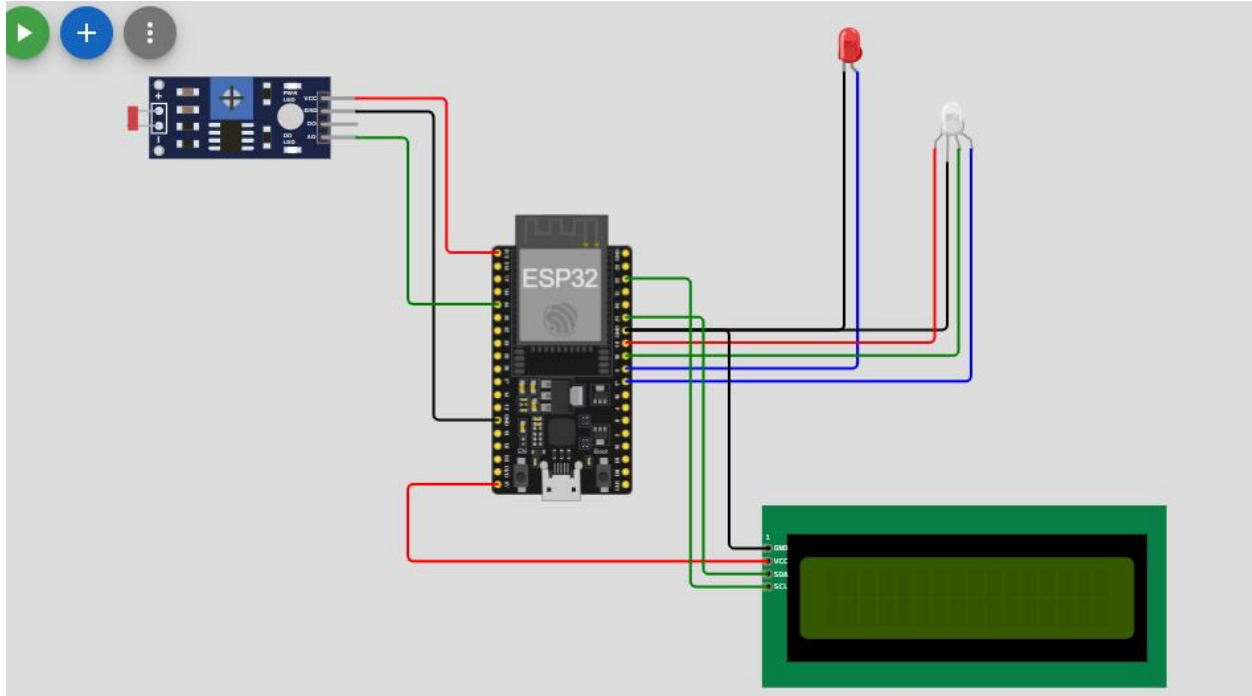
UCT41042 - PRACTICAL FOR INTRODUCTION TO SMART SYSTEMS

CA-03

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Diagram



Code:

```
#include <Arduino.h>
#include <WiFi.h>
#include <ESPSupabase.h>
#include <LiquidCrystal_I2C.h>
#include <time.h>

#define WIFI_SSID "Wokwi-GUEST"
#define WIFI_PASSWORD ""

#define SUPABASE_URL "https://mnyjdwffbiqwiznupecp.supabase.co"
#define SUPABASE_API_KEY
"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOiJzdXBhYmFzZSIsInJlZiI6Im1ueWpkd2ZmYmlxd2l6bnVwZW50Iiwicm9sZSI6ImFub24iLCJpYXQiOiJlODIyODEyODE3MDQsImV4cCI6ImA5Nzg1NzZwNH0.ioUkA_sK996QbTqY0cRmir0Pvr_kxnHPHDoh9BwJd20"
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2);

#define LDR_PIN 34
#define RED_PIN 19
#define GREEN_PIN 18
#define BLUE_PIN 17
#define STREET_LED_PIN 5

Supabase supabase;

unsigned long lastSend = 0;
const unsigned long interval = 5000;

bool blinkState = false;

void connectWiFi();
void setRGB(bool r, bool g, bool b);
void syncTime();
int sendToSupabase(String jsonStr);

void setup()
{
  Serial.begin(115200);
  Serial.println("\n=== SMART STREET LIGHT SYSTEM STARTED ===");

  pinMode(RED_PIN, OUTPUT);
  pinMode(GREEN_PIN, OUTPUT);
  pinMode(BLUE_PIN, OUTPUT);
  pinMode(STREET_LED_PIN, OUTPUT);

  lcd.init();
  lcd.backlight();
  lcd.print("SMART STREET");
  lcd.setCursor(0, 1);
  lcd.print("LIGHT SYSTEM");
  delay(2000);
  lcd.clear();

  connectWiFi();

  // Sync time before any HTTPS/TLS calls
  syncTime();

  // Initialize Supabase
```

```

    supabase.begin(SUPABASE_URL, SUPABASE_API_KEY);
}

void loop()
{
    int ldrValue = analogRead(LDR_PIN);
    int intensity = map(ldrValue, 0, 4095, 0, 100);
    intensity = constrain(intensity, 0, 100);

    String status = "OFF";
    String ledState = "UNKNOWN";

    if (intensity > 70) {
        setRGB(false, true, false);
        digitalWrite(STREET_LED_PIN, LOW);
        ledState = "Green";
    }
    else if (intensity >= 30) {
        setRGB(false, false, true);
        digitalWrite(STREET_LED_PIN, LOW);
        ledState = "BLUE";
    }
    else if (intensity >= 15) {
        setRGB(true, false, false);
        digitalWrite(STREET_LED_PIN, HIGH);
        ledState = "RED";
        status = "ON";
    }
    else {
        blinkState = !blinkState;
        setRGB(blinkState, false, false);
        digitalWrite(STREET_LED_PIN, HIGH);
        status = "ON";
        ledState = "RED_BLINK";
    }

    lcd.setCursor(0, 0);
    lcd.print("Light: ");
    lcd.print(intensity);
    lcd.print("%   ");

    lcd.setCursor(0, 1);
    lcd.print("Street: ");
    lcd.print(status);
    lcd.print("   ");

```

```

    if (millis() - lastSend >= interval) {
        lastSend = millis();

        // Build JSON payload
        JsonDocument doc;
        doc["intensity"] = intensity;
        doc["status"] = status;
        doc["ledState"] = ledState;

        String jsonStr;
        serializeJson(doc, jsonStr);

        int responseCode = sendToSupabase(jsonStr);
        Serial.print("Supabase Response Code: ");
        Serial.println(responseCode);
    }
}

int sendToSupabase(String jsonStr) {
    int responseCode = -1;
    const int maxAttempts = 3;

    for (int attempt = 1; attempt <= maxAttempts; attempt++) {
        responseCode = supabase.insert("street_light_logs", jsonStr, false);

        if (responseCode > 0) {
            // (HTTP 2xx/etc) – stop retrying
            break;
        }

        Serial.print("Insert attempt ");
        Serial.print(attempt);
        Serial.print(" failed (code ");
        Serial.print(responseCode);
        Serial.println("), retrying...");
        delay(1000);
    }

    return responseCode;
}

void connectWiFi() {
    WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
    Serial.print("Connecting to WiFi");

```

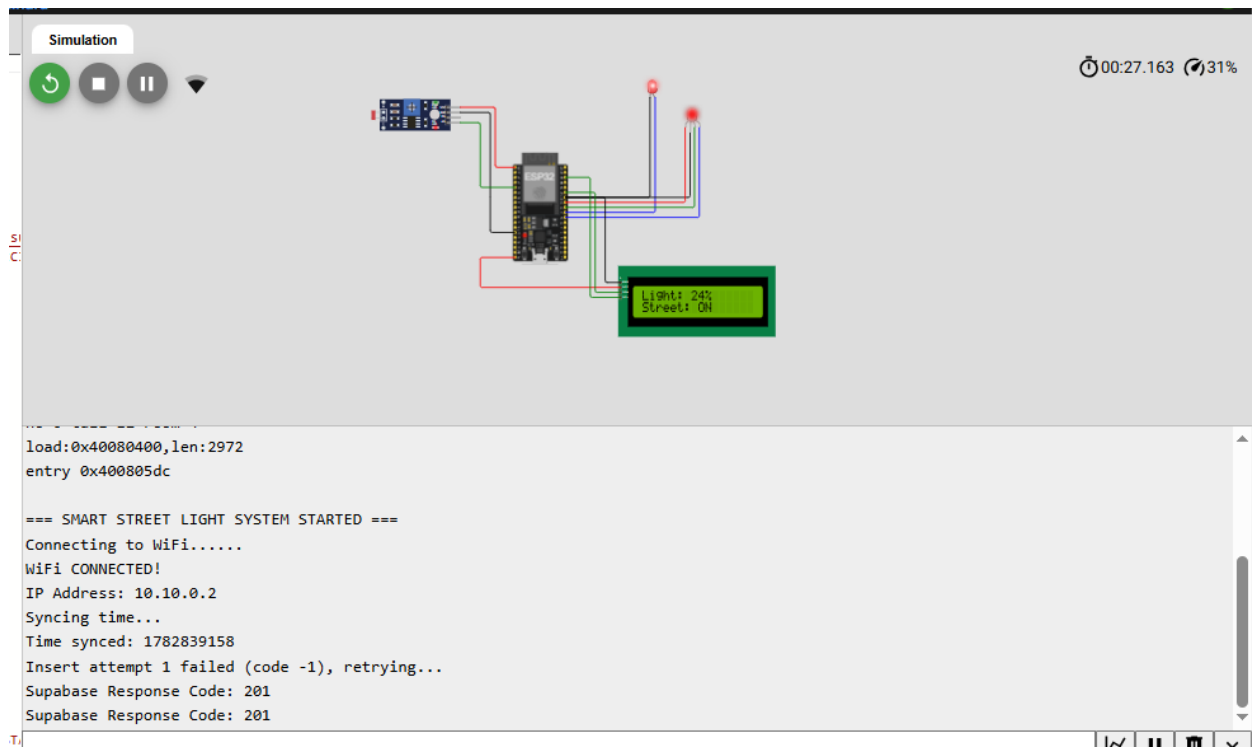
```
while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
}
Serial.println("\nWiFi CONNECTED!");
Serial.print("IP Address: ");
Serial.println(WiFi.localIP());
}

void syncTime() {
    configTime(0, 0, "pool.ntp.org", "time.nist.gov");

    Serial.print("Syncing time");
    time_t now = time(nullptr);
    while (now < 1700000000) {
        delay(500);
        Serial.print(".");
        now = time(nullptr);
    }
    Serial.println("\nTime synced: " + String(now));
}

void setRGB(bool r, bool g, bool b) {
    digitalWrite(RED_PIN, r);
    digitalWrite(GREEN_PIN, g);
    digitalWrite(BLUE_PIN, b);
}
```

Output:



Database:

Keloha / espdb / main (PRODUCTION) / Connect

Feedback Search... Ctrl K

Table Editor

schema public

+ New table

Search tables...

street_light_logs

Filter by id, intensity, status... or ask AI

Sort RLS policy Role postgres

Insert

	id int8	intensity int4	status varchar	ledState varchar	created_at timestamp	
☐	37	57	OFF	BLUE	2026-06-29 13:17:20.448629	
☐	38	24	ON	RED	2026-06-29 15:02:39.783275	
☐	39	24	ON	RED	2026-06-29 15:02:41.566884	
☐	40	49	OFF	BLUE	2026-06-29 15:02:59.688988	
☐	41	69	OFF	BLUE	2026-06-29 15:03:23.164675	
☐	42	3	ON	RED_BLINK	2026-06-29 15:03:45.046911	
☐	43	3	ON	RED_BLINK	2026-06-29 15:04:07.609944	
☐	44	3	ON	RED_BLINK	2026-06-29 15:04:31.759654	
☐	45	3	ON	RED_BLINK	2026-06-29 15:04:52.150484	
☐	46	3	ON	RED_BLINK	2026-06-29 15:05:12.9233	
☐	47	3	ON	RED_BLINK	2026-06-29 15:05:34.702071	
☐	48	3	ON	RED_BLINK	2026-06-29 15:05:58.000990	
☐	49	3	ON	RED_BLINK	2026-06-29 15:06:15.657592	
☐	50	3	ON	RED_BLINK	2026-06-29 15:06:40.738677	
☐	51	24	ON	RED	2026-06-29 15:21:28.457097	

Page 1 of 1 100 rows 95 records

Data Definition

Mobile App:

